NEWTON SCHOOL

MATH · HOMEWORK · NOTES 1 OF 2

Estimate, Multiply & Compare

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4.NBT.B.5 · Multiply two two-digit numbers using place value and an area model.

Goal: Make two estimates, find the exact product with four area blocks, and compare each estimate with the exact answer.

Two ways to estimate

1. Round each factor to the nearest ten

Choose the closer multiple of ten. Halfway? Choose the greater ten. A multiple of ten stays the same.

For 76 × 19: 76 → 80 and 19 → 20.
76 × 19 ≈ 80 × 20 = 1,600.

2. Choose compatible factors

Choose nearby numbers that are easy to multiply together. They do not have to be the nearest tens. You may keep a factor unchanged.

For 76 × 19, choose 75 and 20.
76 × 19 ≈ 75 × 20 = 1,500.
Think: 75 × 2 = 150, then 150 × 10 = 1,500.

Predict before finding the exact product

These rules work when **all factors are positive**.

How the factors change	What you can predict
Both increase	Overestimate
Both decrease	Underestimate
One stays the same	Follow the changed factor: up → over; down → under.
Both stay the same	Exact
One increases, one decreases	Cannot tell from direction alone. Calculate and compare.

> **Why?** Increasing a positive factor makes the product greater. Decreasing a positive factor makes it smaller. With mixed changes, the effects compete.

Worked example · 76 × 19

Find the exact product with four area blocks

Split each factor into tens and ones: **76 = 70 + 6** and **19 = 10 + 9**. Label every block and add all four partial products.

×	10	9
70	70 × 10 = 700	70 × 9 = 630
6	6 × 10 = 60	6 × 9 = 54

700 + 630 + 60 + 54 = 1,444. So 76 × 19 = 1,444.

Compare both estimates with 1,444

Method	Estimate	Comparison and difference
Nearest ten	80 × 20 = 1,600	Over by 156 1,600 − 1,444 = 156
Compatible factors	75 × 20 = 1,500	Over by 56 1,500 − 1,444 = 56

Rounding increases both positive factors, so we can predict an overestimate. The compatible choice lowers 76 but raises 19: **direction alone cannot tell us**. Comparing with 1,444 shows it is over.

> **How far away is an estimate?**
>
> Subtract the smaller product from the greater product. An exact estimate has a difference of **0**. Here, 1,500 is closer because 56 is less than 156.

Your homework routine

For each problem, use the same order:

1. Round both factors to the nearest ten and multiply.
2. Choose nearby compatible factors and multiply.
3. Predict over, under, or exact when the factor directions make it clear. For mixed changes, write “cannot tell.”
4. Draw and label four area blocks. Add the four products.
5. Compare each estimate with the exact product and find each difference.

1. 28 × 37

A. Round each factor to the nearest ten.

28 → _____ 37 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

2. 42 × 63

A. Round each factor to the nearest ten.

42 → _____ 63 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

3. 47 × 58

A. Round each factor to the nearest ten.

47 → _____ 58 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

4. 31 × 74

A. Round each factor to the nearest ten.

31 → _____ 74 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

5. 59 × 22

A. Round each factor to the nearest ten.

59 → _____ 22 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

6. 53 × 39

A. Round each factor to the nearest ten.

53 → _____ 39 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

7. 68 × 41

A. Round each factor to the nearest ten.

68 → _____ 41 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

8. 34 × 27

A. Round each factor to the nearest ten.

34 → _____ 27 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

9. 62 × 48

A. Round each factor to the nearest ten.

62 → _____ 48 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

10. 73 × 26

A. Round each factor to the nearest ten.

73 → _____ 26 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

11. 24 × 38

A. Round each factor to the nearest ten.

24 → _____ 38 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

12. 49 × 26

A. Round each factor to the nearest ten.

49 → _____ 26 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

13. 74 × 19

A. Round each factor to the nearest ten.

74 → _____ 19 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

14. 51 × 24

A. Round each factor to the nearest ten.

51 → _____ 24 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

15. 38 × 23

A. Round each factor to the nearest ten.

38 → _____ 23 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

16. 87 × 26

A. Round each factor to the nearest ten.

87 → _____ 26 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

17. 52 × 48

A. Round each factor to the nearest ten.

52 → _____ 48 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

18. 63 × 24

A. Round each factor to the nearest ten.

63 → _____ 24 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

19. 95 × 32

A. Round each factor to the nearest ten.

95 → _____ 32 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____

20. 67 × 45

A. Round each factor to the nearest ten.

67 → _____ 45 → _____ Estimate: _____ × _____ = _____

Before exact work: over / under / exact / cannot tell. Why? _____

B. Choose nearby compatible factors.

Estimate: _____ × _____ = _____ Why easy? _____

Before exact work: over / under / exact / cannot tell. Why? _____

C. Find the exact product using four area blocks.

Split both factors into tens and ones. Draw four blocks below; label the sides and all four products.

Add all four: _____ + _____ + _____ + _____ = _____

D. Compare each estimate with your exact product.

Estimate	Circle one	Difference (subtract)
Nearest ten	over / under / exact	_____
Compatible	over / under / exact	_____

Which estimate is closer, or are they equally close? _____